ARPITA MALLIK

+880-1552398872 | arpitamallik13@gmail.com | Portfolio - Arpita Mallik

Arpita Mallik | Arpita Mallik | arpitamallik23

Chittagong, Bangladesh

Summary

I am a recent Computer Science and Engineering graduate. My research interests include Large Language Models, Natural Language Processing, Multilingual and Low-resource NLP, Parameter-Efficient Fine-tuning, and Trustworthy AI, especially in terms of factuality of LLMs, their sycophantic tendencies, deception, and safety concerns.

Education

- **Chittagong University of Engineering & Technology** *March 2022 – July 2026*
  BSc. in Computer Science & Engineering Chittagong, Bangladesh
  - GPA: 3.63/4.00 (most recent 78.25 credits: 3.87/4.00)

Test Scores

- **TOEFL iBT** *17 October 2026*
  Overall: 6.0/6 (118/120), CEFR C2
  - Reading: 6.0 Listening: 5.0 Writing: 6.0 Speaking: 6.0

Research Experience

- **Research Intern, ELITE Research Lab** *June 2026 – Present*
  Adv. Md Kishor Morol - Founder and CEO, Elite Research Lab
  - Building a counterfactual multimodal deception benchmark for vision-language models.

  Machine Learning Instructor *April 2025 – April 2026*
  Andromeda Space and Robotics Research Organisation, CUET
  - Led and contributed to multiple competitive shared tasks across multimodal misogyny detection and Bengali polarization detection.
  - Mentored multiple research projects, contributing to workshop papers accepted at ACL Anthology venues such as DravidianLangTech and LREC.

Thesis

- **Undergraduate Thesis** *2026 – Present*
  An Emotion-Aware Large Language Model-based QA System for Women's Healthcare Information in Bangla
  - Curated and publicly released MAYA, a Bengali and Banglish women's health QA dataset (3,002 pairs) with neutral/concerned emotion labels across eight topics, including menstrual health, PCOS, pregnancy, menopause and postpartum depression.
  - Built an emotion-aware QA system by fine-tuning Gemma-2-9B with QLoRA on 2,500 pairs, adding FiLM emotion conditioning via forward hooks and a BanglaBERT classifier (94.82% emotion accuracy). Test BERTScore F1 reached 87.82%.
  - Showed BERTScore is a weak quality signal across seven fine-tuned models (Gemma, Llama, Mistral, Qwen, Phi): within each model it explained only 6–16% of the variation in GPT-judged correctness and 1–7% in safety. 5–45% of answers above 0.85 BERTScore were judged wrong or unsafe.
  - Compared zero-shot, LoRA-only, and FiLM+LoRA systems. FiLM matched LoRA on automatic metrics, and a 50-sample human evaluation found plain prompting gave better tone control than FiLM.
  - Ran an LLM-as-judge sycophancy probe on 102 health myths (Bangla/Banglish, neutral/concerned framing). Fine-tuning raised judge scores from 0.966 to 0.984, but FiLM vs LoRA showed no significant difference.

Publications

C=Conference, J=Journal, S=In Submission, T=Thesis

[C.1] **CUET-823@DravidianLangTech 2025: Shared Task on Multimodal Misogyny Meme Detection in Tamil Language**. In *Proceedings of the Fifth Workshop on Speech, Vision, and Language Technologies for Dravidian Languages.*

[C.2] **CUET-823 at SemEval-2026 Task 9: LoRA-Based Instruction-Fine-Tuned LLMs vs. Transformer Models for Bengali Polarization Detection**. Accepted to the *20th International Workshop on Semantic Evaluation (SemEval-2026)*, 2026.

Currently in Progress

- **Counterfactual Multimodal Deception Benchmark** *2026 – Present*
  ELITE Research Lab
  - Designing each scenario against a frozen 2×4 schema (Image_Type: Original/Visual_CF × Prompt_Type: Baseline/Incentive/Oversight/Correction)
  - Building out the dataset across Food Safety, Safety & Security, Wildlife & Nature, and Traffic & Transportation domains.
- **Survey on Trustworthy Explainable AI (XAI) in Oncology** *2026 – Present*
  - Co-developing a survey on trustworthy XAI for oncology and clinical decision support, examining fairness and equity, explanation robustness, security and vulnerabilities, explanation validity, and clinical validity.
  - Co-proposing a multidimensional evaluation framework that organizes trustworthy XAI around three core dimensions: fairness, robustness, and security, with explanation validity as a prerequisite gate covering faithfulness, context awareness, and sanity checks.

Under Review

[P.1] **Gap-Aware Windowing and Task-Specific Preprocessing for Bengali Long-Form ASR and Diarization**. *ICCIT 2026*

[P.2] **Assessing the Reliability of Large Language Models in the Bengali Legal Context: A Comparative Evaluation Using LLM-as-Judge and Legal Experts**. *ICCIT 2026*

Teaching Experience

- **ASSRO, CUET** *2025 – 2026*
  Machine Learning Instructor Chittagong, Bangladesh
  - Mentored junior students in Machine Learning and Deep Learning through technical sessions and hands-on workshops.
  - Guided research and implementation for NLP, Computer Vision and Multimodal shared-task projects, contributing to workshop paper publications at ACL Anthology venues.

Technical Skills

- **Programming Languages:** Python, SQL
- **ML/DL Frameworks:** PyTorch, scikit-learn
- **NLP & LLMs:** Transformers, Hugging Face, QLoRA, LoRA, RAG
- **Deployment & Tools:** FastAPI, Docker, AWS EC2/S3, Git, GitHub Actions, MLflow, DVC

Awards & Honors

Finalist – DL Sprint 4.0 (Top 10% / 192 teams, nationwide) *February 2026*
BUET CSE Fest 2026

- Built a pipeline for Bangla long-form speech recognition and speaker diarization using Whisper-based ASR and fine-tuned diarization models.

Champion – NodeScape Hackathon (1st place) *July 2025*
CUET Computer Club

- Developed an AI-powered graph classification system with algorithm visualization; Dockerized CI/CD deployment using React, Flask, and Python.

Finalist – Intra CUET Machine Learning Contest *July 2025*
IEEE CS CUET Student Branch Chapter

- Built a Bangla physics MCQ question-answering system leveraging LLMs for concept-based queries.

Top 7 Finalist – HackCSB 2024 (out of 116 teams) *August 2024*
Coding Shikhbe Bangladesh

- Built "MotivationMate," a mental health-focused web application.

Leadership & Service

Vice Chair (Machine Learning Wing) *July 2025 – Present*
IEEE CS CUET Student Branch Chapter